

Design and Verification of a 16-bit Three-Stage Pipelined ALU with RAW Data Forwarding

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Abstract—This work presents a 16-bit three-stage pipelined Arithmetic Logic Unit (ALU) implemented in Verilog HDL to demonstrate instruction-level overlap in a compact digital datapath. The design is motivated by the need to reduce the effective execution time of repeated arithmetic and logical operations compared with a non-overlapped datapath, while keeping the architecture simple enough for RTL-level verification and FPGA-oriented study. The proposed datapath divides instruction processing into Decode, Execute, and Writeback stages separated by ID/EX and EX/WB pipeline registers. Stage 1 decodes a 16-bit instruction, reads operands from an 8×16 -bit register file, and forwards the decoded fields to the next stage. Stage 2 implements a 16-bit ALU supporting addition, subtraction, bitwise AND/OR/XOR, NOT, logical shifts, and comparison, together with Zero, Carry, Overflow, and Negative status flags. Stage 3 commits the execution result to the destination register. Basic read-after-write (RAW) forwarding is included to support dependent instructions without unnecessarily waiting for register writeback. A self-checking Verilog testbench is used to verify representative arithmetic and logical instruction sequences and to generate waveform data for timing inspection. For a five-instruction sequence, an analytical three-stage pipeline schedule requires seven cycles after pipeline overlap, compared with fifteen cycles for a non-overlapped three-cycle-per-instruction baseline, giving a $2.14\times$ cycle-count reduction for that sequence. The implementation establishes a baseline for future hazard handling, synthesis, FPGA measurement, and extension toward a larger pipelined processor.

Index Terms—pipelining, 16-bit ALU, Verilog HDL, RTL, datapath, pipeline register, RAW hazard, data forwarding, FPGA

I. CONTRIBUTIONS

The main contributions of this work are as follows:

- 1) **Three-stage 16-bit datapath.** A compact pipeline is divided into Decode, Execute, and Writeback stages with explicit ID/EX and EX/WB registers.
- 2) **Multi-operation ALU.** The Execute stage supports ADD, SUB, AND, OR, XOR, NOT, logical shift-left, logical shift-right, and compare operations, with Zero, Carry, Overflow, and Negative flags.
- 3) **RAW data forwarding.** Basic forwarding paths are included so that a dependent instruction can use a recently generated result without waiting for normal register writeback.
- 4) **RTL verification flow.** A self-checking Verilog testbench verifies representative instruction sequences,

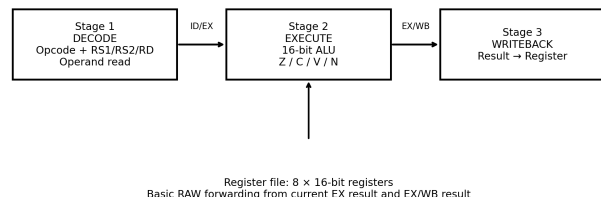


Fig. 1. Three-stage 16-bit pipelined ALU architecture with ID/EX and EX/WB interfaces.

checks final register values, prints pass/fail status, and produces a VCD file for waveform analysis.

II. RESULTS

A. Experimental setup

The design is written in synthesizable Verilog HDL and organized as a three-stage synchronous datapath. The baseline register file contains eight 16-bit registers. Instructions use opcode in bits [15:12], source register 1 in [11:9], source register 2 in [8:6], destination register in [5:3], and three reserved bits in [2:0]. Functional simulation is intended to be performed with Icarus Verilog in EDA Playground. The complete testbench initializes registers, applies a sequence of dependent and independent instructions, checks final register values, and generates a VCD waveform. Pipelining and hazard-resolution approaches follow established datapath concepts used in pipelined processor studies.

Figure 1 summarizes the implemented architecture. The Decode stage prepares operands and control information; the Execute stage performs the ALU operation; and the Writeback stage commits the result to the register file. The organization is intentionally smaller than a conventional five-stage processor, allowing the complete datapath and verification flow to be demonstrated within a compact project.

B. Pipeline behavior

Figure 2 shows the intended overlap for five back-to-back instructions. After the pipeline is filled, a new instruction can

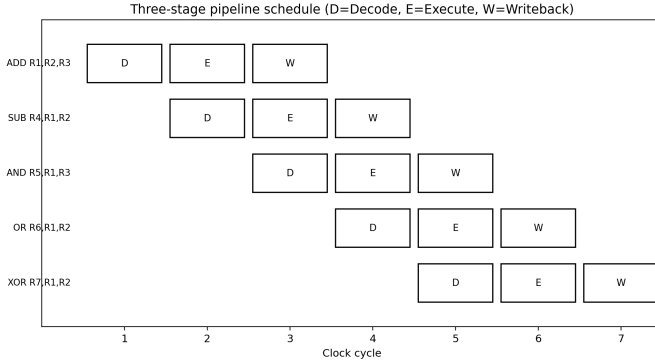


Fig. 2. Three-stage pipeline schedule for five representative instructions.

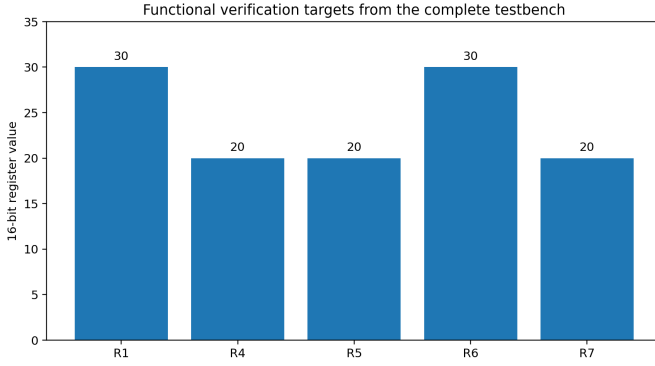


Fig. 3. Expected final register values for the self-checking verification sequence.

enter Decode on each clock cycle while earlier instructions occupy Execute and Writeback. This figure is a timing illustration of the RTL stage organization, not a post-synthesis timing measurement.

C. Functional verification

The verification sequence uses initial values $R2 = 10$ and $R3 = 20$:

```
ADD R1, R2, R3
SUB R4, R1, R2
AND R5, R1, R3
OR R6, R1, R2
XOR R7, R1, R2
```

The expected results are $R1 = 30$, $R4 = 20$, $R5 = 20$, $R6 = 30$, and $R7 = 20$. The second instruction is intentionally dependent on the first so that the forwarding path is exercised. Figure 3 summarizes the values checked by the self-checking testbench. The values should be replaced by captured simulator output if the final experimental instruction sequence is changed.

D. Cycle-count analysis

For a five-instruction sequence, the three-stage pipeline completes the overlapped schedule in seven clock cycles: three cycles fill the pipeline and four additional cycles complete

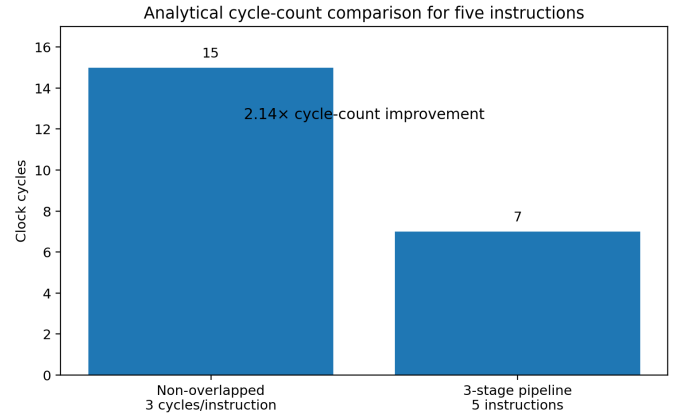


Fig. 4. Analytical cycle-count comparison for five instructions.

the remaining instructions. A simple non-overlapped baseline requiring three cycles per instruction would require fifteen cycles. Therefore,

$$G = \frac{15}{7} = 2.14. \quad (1)$$

Figure 4 shows this analytical comparison. The value is a cycle-count comparison, not a measured FPGA speedup. Actual maximum frequency, area, power, and long-stream throughput must be obtained after synthesis and timing analysis.

E. Discussion

The results establish a functional baseline for a larger pipelined processor datapath. The current implementation demonstrates stage overlap and basic RAW forwarding. The evaluation is limited to RTL simulation and analytical cycle counting. Future work should include synthesis on a selected FPGA, post-synthesis maximum-frequency measurement, LUT/flip-flop utilization, power estimation, more complete hazard detection and stall logic, and extension of the instruction set if the design is expanded toward a processor core.

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